

Intermittent Discrete Observation Control for Synchronization of Stochastic Neural Networks

Yongbao Wu, Jilin Zhu, and Wenxue Li 

Abstract—In this paper, to investigate the exponential synchronization of stochastic neural networks, a new periodically intermittent discrete observation control (PIDOC) is first proposed. Different from the existing periodically intermittent control, our control in control time is feedback control based on discrete-time state observations (FCDSOs) instead of a continuous-time one. By employing the Lyapunov method, graph theory, and theory of differential inclusions, the exponential synchronization of stochastic neural networks with a discontinuous right-hand side is realized by PIDOC and some sufficient conditions are presented. Especially, when control width tends to control period, PIDOC will be reduced to a general FCDSO and we give some detailed discussions. Then, we provide some corollaries about synchronization in mean square, asymptotical synchronization in mean square, and exponential synchronization of stochastic neural networks under FCDSO. Finally, some numerical simulations are provided to demonstrate our analytical results.

Index Terms—Discrete-time state observations, exponential synchronization, periodically intermittent control (PIC), stochastic neural networks.

I. INTRODUCTION

RECENTLY, neural networks have been widely investigated for their fruitful applications in different fields, such as brain-like emulation, knowledge acquisition, combinatorial optimization, pattern recognition, etc. [1]–[6]. Due to extensive applications of neural networks, it is essential to conduct some deep research on neural networks. It is necessary to discuss the dynamical properties of neural networks, such as stability, synchronization, and so on [7]–[11].

Especially, as a typical collective dynamical property of neural networks, synchronization is a hot topic due to its comprehensive applications in a wide variety of fields, including secure communication, information science, image processing, harmonic oscillation generation, etc. [12]–[19]. Recently,

some literature of synchronization and stability for neural networks has been reported [7], [20]–[25]. For example, Zhang and Shen [23] discussed the exponential synchronization of memristive chaotic neural networks with time-varying delays and general activation functions. Based on the aforementioned literature, it is worth noting that most of the above literature ignored stochastic disturbance in neural networks. However, neural networks with stochastic disturbance are more general in practice [26]. For example, in a real nervous system, the synaptic transmission is a process with noises caused by random probabilistic reasons from the release of neurotransmitters and other probabilistic reasons. Due to the influence of stochastic disturbance on neural networks, stochastic disturbance should be considered in this paper.

Many scholars investigated the dynamical properties of complex networks by sampling control [27], [28] or intermittent control [29]–[36]. For example, by employing periodically intermittent control (PIC), Zhang and Shen [23] obtained several new criteria for exponential synchronization of two memristive chaotic neural networks. Then, Yang *et al.* [36] investigated the issue of exponential synchronization for the fuzzy model of neural networks by employing PIC. According to the above literature, we know that this control is very effective. Moreover, in the real world, PIC is easier to implement because it has nonzero control time. However, it is worth noting that PICs in most of the above literature are based on continuous-time state observations in the entire control time. In fact, recently, Mao [37] presented a new feedback control based on discrete-time state observations (FCDSOs) to discuss mean-square exponential stabilization of continuous-time hybrid stochastic differential equations. This new control only observes the discrete-time state. From the perspective of control, when taking the cost and efficiency into consideration, observing discrete-time state is more cost effective and practical than continuous-time state. So many researchers have employed FCDSO to study the dynamical properties of stochastic systems [38]–[40]. Although there exist interesting results in many kinds of stochastic systems, such as hybrid stochastic differential equations, stochastic nonlinear systems, and stochastic coupled systems, few papers used FCDSO to study the exponential synchronization of stochastic neural networks. Moreover, most of the above FCDSOs are needed all of the time which may be non-economic. Considering the advantages of PIC, we first propose a novel control called periodically intermittent discrete observation control (PIDOC), which only observes discrete-time state in control time instead of continuous-time state. In this

Manuscript received February 17, 2019; revised June 20, 2019; accepted July 18, 2019. Date of publication August 7, 2019; date of current version May 7, 2020. This work was supported in part by the Shandong Province Natural Science Foundation under Grant ZR2018MA005, Grant ZR2018MA020, and Grant ZR2017MA008, in part by the Key Project of Science and Technology of Weihai under Grant 2014DXGJMS08, and in part by the Innovation Technology Funding Project in Harbin Institute of Technology under Grant HIT.NSRIF.201703. This paper was recommended by Associate Editor Z. Zeng. (Corresponding author: Wenxue Li.)

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Digital Object Identifier 10.1109/TCYB.2019.2930579

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paper, we first try to use PIDOC to study the exponential synchronization of stochastic neural networks. That is our motivation.

Motivated by the above analysis, in this paper, we propose a PIDOC to study the exponential synchronization of stochastic neural networks. PIDOC has the advantages of PIC and FCDSO. Then, the Lyapunov method, graph theory, and theory of differential inclusions are employed to obtain some synchronization criteria of stochastic neural networks. Finally, some numerical simulations are displayed to substantiate our theoretical results. Compared with the existing results, the main contributions of this paper are listed as follows.

- 1) Different from PIC and FCDSO in [23] and [36]–[40], in this paper, we design a novel PIDOC to investigate the exponential synchronization of stochastic neural networks.
- 2) The exponential synchronization of stochastic neural networks with discontinuous right-hand side is realized by PIDOC. Moreover, when control width tends to control period, PIDOC becomes FCDSO. Compared with the previous work, PIDOC is more general and appropriate in practice.
- 3) By employing the Lyapunov method, graph theory, and theory of differential inclusions, some new synchronization criteria are obtained for stochastic neural networks.

The remainder of this paper is organized as follows. In Section II, we introduce some model formulation and preliminaries. Some sufficient conditions and some corollaries are obtained in Section III. Then, in Section IV, some numerical simulations are given. Eventually, in Section V, we give the conclusion.

Notations: The following notations will be used throughout this paper. First, let $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ be a complete probability space with a filtration $\mathbb{F} = \{\mathcal{F}_t\}_{t \geq 0}$ satisfying the usual conditions (i.e., it is right continuous and $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets). The mathematical expectation with respect to the given probability measure $\mathbb{P}$ is denoted by $\mathbb{E}(\cdot)$. Superscript “T” stands for the transpose of a vector. Let $\mathbb{R}$ denote the set of real numbers and $\mathbb{R}^n$ be the space of n -dimensional Euclidean space and denote $|\cdot|$ for the Euclidean norm in $\mathbb{R}^m$. Let $\mathcal{N} = \{1, 2, \dots, n\}$, $\mathbb{Z}^+ = \{1, 2, \dots\}$, $\mathbb{R}^+ = [0, \infty)$.

We define $\mathcal{G} = (v, \varepsilon, A)$ as a weighted digraph with the set of nodes $v = (v_1, v_2, \dots, v_n)$, set of edges $\varepsilon \subseteq v \times v$, and a weighted adjacency matrix $A = (a_{ij})_{n \times n}$, in which adjacency elements a_{ij} are non-negative. The node indices belong to a finite index set $\mathcal{N}$. Denote an edge by $a_{ij} = (v_i, v_j)$. It is worth noting that every adjacency element related to the edges is positive. Let $a_{ii} = 0$ for all $i \in \mathcal{N}$. In a digraph, if any two distinct nodes could be connected by a path that follows the direction of the edges, this digraph is defined as strongly connected. The Laplacian matrix of digraph $(\mathcal{G}, A)$ is defined by $L = (l_{kh})_{n \times n}$, where $l_{kh} = -a_{kh}$, for $k \neq h$, $l_{kh} = -\sum_{h \neq k} a_{kh}$, for $k = h$.

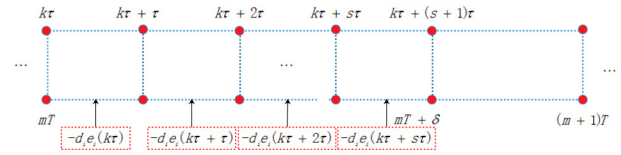


Fig. 1. Sketch map of PIDOC, $e_i(t) = y_i(t) - x_i(t)$.

II. MODEL FORMULATION AND PRELIMINARIES

In this section, we consider the following stochastic neural networks that:

$$dx_i(t) = \left(-x_i(t) + \sum_{j=1}^n a_{ij}(x_j(t)) f_j(x_j(t)) + I_i \right) dt + g_i(x_i(t), t) dB(t), \quad t \geq 0, \quad i \in \mathcal{N} \quad (1)$$

where $x_i(t) \in \mathbb{R}$ and

$$a_{ij}(x_j(t)) = \begin{cases} a_{ij}^*, & |x_j(t)| \leq T_j, \\ a_{ij}^{**}, & |x_j(t)| > T_j, \end{cases} \quad t \geq 0, \quad i, j \in \mathcal{N}$$

in which switching jumps $T_j > 0$, a_{ij}^* , and a_{ij}^{**} are some constants. Moreover, $f_j : \mathbb{R} \rightarrow \mathbb{R}$ denotes the neuron activation function and $g_i : \mathbb{R} \rightarrow \mathbb{R}$ denotes the noise intensity. And $B(t)$ is 1-D Brownian motion defined on the probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$. I_i is an external constant input.

Throughout this paper, we consider system (1) as a drive system, and the corresponding response system with PIDOC is given as follows:

$$dy_i(t) = \left(-y_i(t) + \sum_{j=1}^n a_{ij}(y_j(t)) f_j(y_j(t)) + I_i + u_i(t) \right) dt + g_i(y_i(t), t) dB(t), \quad t \geq 0, \quad i \in \mathcal{N} \quad (2)$$

where

$$a_{ij}(y_j(t)) = \begin{cases} a_{ij}^*, & |y_j(t)| \leq T_j, \\ a_{ij}^{**}, & |y_j(t)| > T_j, \end{cases} \quad t \geq 0, \quad i, j \in \mathcal{N}$$

and $u_i(t)$ is PIDOC which is designed as

$$u_i(t) = \begin{cases} -d_i(y_i(\delta(t)) - x_i(\delta(t))), & mT \leq t \leq mT + \delta \\ 0, & mT + \delta < t < (m+1)T \end{cases} \quad (3)$$

where $m = 0, 1, \dots$, and d_i is a positive constant called the control gain. Moreover, $\delta(t) = [t/\tau]\tau$, where $\tau > 0$ is the duration between two consecutive state observations, and $[t/\tau]$ is the integer of t/τ . Meanwhile, we assume that $\delta/\tau \in \mathbb{Z}^+$ and $mT = k\tau$ for $m \in \mathbb{Z}^+$. $[mT, mT + \delta]$ is the control time, while $(mT + \delta, (m+1)T)$ is the rest time. T denotes the control period and $0 < \delta < T$. Then, Fig. 1 is given to show the control strategy more clearly.

Remark 1: In this paper, a new PIDOC is proposed. In particular, when $\tau \rightarrow 0$, $-d_i(y_i(\delta(t)) - x_i(\delta(t)))$ in control time turns to $-d_i(y_i(t) - x_i(t))$, that is, controller (3) becomes an ordinary periodically intermittent controller [23], [36]. Meanwhile, when $\delta \rightarrow T$, our control turns into FCDSOs [37]–[39]. Therefore, our control is a generalization of a periodically intermittent controller and FCDSOs.

Remark 2: In this paper, the control function in control time is designed as $-d_i(y_i(\delta(t)) - x_i(\delta(t)))$, where $\delta(t) = [t/\tau]\tau$ and $[t/\tau]$ is the integer of t/τ . In fact, if $m\tau \leq t < (m+1)\tau$, then $[t/\tau]\tau = m\tau$, where $m = 0, 1, 2, \dots$. Thus, when $m\tau \leq t < (m+1)\tau$, we know $-d_i(y_i(\delta(t)) - x_i(\delta(t))) = -d_i(y_i(m\tau) - x_i(m\tau))$. In general, continuous-time feedback control requires the continuous-time state observations of the state vector $y_i(t) - x_i(t)$ for all $t \geq 0$. However, the feedback control $-d_i(y_i(\delta(t)) - x_i(\delta(t)))$ only needs the discrete-time state observations of the state vector $y_i(t) - x_i(t)$ at time $0, \tau, 2\tau, \dots$, where τ is the duration between two consecutive state observations. Therefore, it is more reasonable to adopt FCDSOs in control time, and its cost will be less than feedback control based on the continuous-time state observations.

Remark 3: The PIDOC is designed to study the synchronization of stochastic neural networks in this paper. We require that $\delta/\tau \in \mathbb{Z}^+$. If $\delta/\tau \notin \mathbb{Z}^+$, or PIDOC turns to aperiodically intermittent discrete observation control, then it would be an interesting issue for the synchronization of stochastic neural networks under the new control. In the future, we will consider the synchronization of stochastic neural networks in this case.

Definition 1 [41]: The Filippov set-valued map of $f(x)$ at $x \in \mathbb{R}^n$ is defined as follows:

$$F(x) = \bigcap_{\varepsilon > 0} \bigcap_{\mu(\mathcal{D})=0} \overline{\text{co}}[f(B(x, \varepsilon) \setminus \mathcal{D})]$$

where $B(x, \varepsilon) = \{y : |y - x| \geq \varepsilon\}$, and $\mu(\mathcal{D})$ is the Lebesgue measure of set $\mathcal{D}$.

In this paper, all solutions of the systems considered in the following equations are intended in Filippov's sense [41]. By Definition 1 and the theory of differential inclusions, one can obtain that

$$\begin{aligned} dx_i(t) \in & \left(-x_i(t) + \sum_{j=1}^n \text{co}[a_{ij}(x_j(t))]f_j(x_j(t)) + I_i \right) dt \\ & + g_i(x_i(t), t)dB(t), \quad t \geq 0, \quad i \in \mathcal{N} \end{aligned}$$

and

$$\begin{aligned} dy_i(t) \in & \left(-y_i(t) + \sum_{j=1}^n \text{co}[a_{ij}(y_j(t))]f_j(y_j(t)) + I_i + u_i(t) \right) dt \\ & + g_i(y_i(t), t)dB(t), \quad t \geq 0, \quad i \in \mathcal{N} \end{aligned}$$

where

$$\text{co}[a_{ij}(x_j(t))] = \begin{cases} a_{ij}^*, & |x_j(t)| < T_j \\ \text{co}\{a_{ij}^*, a_{ij}^{**}\}, & |x_j(t)| = T_j \\ a_{ij}^{**}, & |x_j(t)| > T_j \end{cases}$$

or, equivalently, there exist measurable functions $\tilde{a}_{ij}(t) \in \text{co}[a_{ij}(x_j(t))]$ and $\hat{a}_{ij}(t) \in \text{co}[a_{ij}(y_j(t))]$, for a.e. $t \geq 0$, such that

$$\begin{aligned} dx_i(t) = & \left(-x_i(t) + \sum_{j=1}^n \tilde{a}_{ij}(t)f_j(x_j(t)) + I_i \right) dt \\ & + g_i(x_i(t), t)dB(t), \quad i \in \mathcal{N} \end{aligned} \quad (4)$$

$$\begin{aligned} dy_i(t) = & \left(-y_i(t) + \sum_{j=1}^n \hat{a}_{ij}(t)f_j(y_j(t)) + I_i + u_i(t) \right) dt \\ & + g_i(y_i(t), t)dB(t), \quad i \in \mathcal{N}. \end{aligned} \quad (5)$$

Now, define the synchronization error vector as follows: $e_i(t) = y_i(t) - x_i(t)$ for $i \in \mathcal{N}$. Combining (4) with (5), we obtain

$$\begin{aligned} de_i(t) = & \left(-e_i(t) + \sum_{j=1}^n \hat{a}_{ij}(t)f_j(y_j(t)) \right. \\ & \left. - \sum_{j=1}^n \tilde{a}_{ij}(t)f_j(x_j(t)) + u_i(t) \right) dt \\ & + (g_i(y_i(t), t) - g_i(x_i(t), t))dB(t), \quad i \in \mathcal{N}. \end{aligned} \quad (6)$$

To derive the main results, two assumptions, two lemmas, and some definitions are given as follows.

Assumption 1: For any $j \in \mathcal{N}$, $s_1, s_2 \in \mathbb{R}$, the activation function f_j is bounded, while $f_j(\pm T_j) = 0$, and satisfies

$$\rho_j^- \leq \frac{f_j(s_1) - f_j(s_2)}{s_1 - s_2} \leq \rho_j^+ \quad (7)$$

where $s_1 \neq s_2$, ρ_j^- , and ρ_j^+ are constants. And there exists a positive constant η_i ensuring that function g_i satisfies the following condition:

$$|g_i(y, t) - g_i(x, t)| \leq \eta_i |y - x|, \quad i \in \mathcal{N}.$$

Remark 4: In fact, condition (7) in Assumption 1 is the generalization of function f_j meeting condition $|f_j(s_1) - f_j(s_2)| \leq \rho_j |s_1 - s_2|$. In (7), we know that ρ_j^- and ρ_j^+ can be negative, zero, or positive. In fact, in many existing results on neural networks, the activation functions satisfy (7). For example, Chua and Yang [42] used the piecewise linear activation function $f_i(x) = (x + 1 - |x - 1|)/2$ in cellular neural networks. In [43], the sigmoid function $f_i(x) = \tanh(x)$ was used in the Hopfield neural networks.

Assumption 2: Digraph $(\mathcal{G}, A)$ is strongly connected in which

$$A = (A_{ij}\rho_j)_{n \times n}$$

where $A_{ij} = \max\{|a_{ij}^*|, |a_{ij}^{**}|\}$ and $\rho_j = \max\{|\rho_j^-|, |\rho_j^+|\}$.

Lemma 1 [44]: Suppose that $n \geq 2$. Let c_i denote the cofactor of the i th diagonal element of the Laplacian matrix of digraph $(\mathcal{G}, A)$. Then, the following identity holds:

$$\sum_{i,j=1}^n c_i a_{ij} F_{ij}(x_i, x_j) = \sum_{\mathcal{Q} \in \mathcal{Q}} W(\mathcal{Q}) \sum_{(s,r) \in E(C_{\mathcal{Q}})} F_{rs}(x_r, x_s)$$

where $F_{ij}(x_i, x_j) : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ are the arbitrary functions, $\mathcal{Q}$ is the set of all spanning unicyclic graphs of digraph $(\mathcal{G}, A)$ that denote the digraph with weight matrix A , $W(\mathcal{Q})$ is the weight of $\mathcal{Q}$, and $C_{\mathcal{Q}}$ denotes the directed cycle of $\mathcal{Q}$. Particularly, if digraph $(\mathcal{G}, A)$ is strongly connected, then $c_i > 0$ for any $i \in \mathcal{N}$.

Lemma 2 [45]: Let $u(\cdot)$ be a Borel measurable-bounded non-negative function on $[0, T]$, and $v(\cdot)$ be a non-negative

integrable function on $[0, T]$. If

$$u(t) \leq c + \int_0^t v(s)u(s)ds, \text{ for all } 0 \leq t \leq T$$

then

$$u(t) \leq c \left(\int_0^t v(s)ds \right), \text{ for all } 0 \leq t \leq T.$$

Definition 2 [45]: Drive system (1) and response system (2) are said to be exponentially synchronous if there exist constants $q > 0$ and $\chi > 0$ such that

$$\mathbb{E}|e(t)|^2 \leq \chi \exp\{-qt\}, \quad t \geq 0$$

where q is called the convergence rate of exponential synchronization and $e(t) = (e_1(t), e_2(t), \dots, e_n(t))^T \in \mathbb{R}^n$.

Definition 3 [46]: Drive system (1) and response system (2) are said to be synchronous in mean square if error system (6) is stable in mean square, that is, for any $e(0) = e_0 \in \mathbb{R}^n$, it is satisfied that $\lim_{t \rightarrow \infty} \mathbb{E} \int_0^t |e(s)|^2 ds < \infty$.

Definition 4 [47]: Drive system (1) and response system (2) are said to be asymptotically synchronous in mean square, if for any initial value $e(0) = e_0 \in \mathbb{R}^n$, then it yields that $\lim_{t \rightarrow \infty} \mathbb{E}|e(t)|^2 = 0$.

III. MAIN RESULTS

In this section, based on the Lyapunov method and graph theory, some exponential synchronization criteria about stochastic neural networks are provided. At first, we introduce some useful notations used in the main results as follows. Set

$$\begin{aligned} d &= \max_{i \in \mathcal{N}} \left\{ \frac{d_i^2}{2\lambda_i} \right\}, \beta = \min_{i \in \mathcal{N}} \{\theta_i\}, \phi = \max_{i \in \mathcal{N}} \{\phi_i\} \\ K(\tau) &= 2 \exp\{\tau\phi\} \max_{i \in \mathcal{N}} \left\{ \tau\phi_i + 6\tau^2 d_i^2 \right\} \\ \phi_i &= 2 \left[3\tau \left(1 + n \max_{i,j \in \mathcal{N}} \{A_{ij}\rho_j\} \sum_{j=1}^n A_{ij}\rho_j \right) + \eta_i^2 \right] \\ \alpha &= \max_{i \in \mathcal{N}} \left\{ \eta_i^2 + \sum_{j=1}^n A_{ij}\rho_j - 2 \right\} \\ \theta_i &= 2 - 2 \sum_{j=1}^n A_{ij}\rho_j - 2\lambda_i + 2d_i - \eta_i^2 - \frac{dK(\tau)}{1 - K(\tau)} \end{aligned}$$

where λ_i is a positive number.

In this section, our main results, including some theorems and corollaries, are displayed as follows.

Theorem 1: Under Assumptions 1 and 2, there exists a sufficiently small duration τ between two consecutive state observations such that $k(\tau) < 1$. Then, drive system (1) and response system (2) reach exponential synchronization under controller (3) if the following inequalities are satisfied:

$$\alpha > 0, \quad \beta > 0, \quad \delta\beta - (T - \delta)\alpha > 0.$$

Proof: Let function $V(e, t) = \sum_{i=1}^n c_i V_i(e_i, t)$, where $V_i(e_i, t) = e_i^2$, c_i denotes the cofactor of the i th diagonal element of the Laplacian matrix of digraph $(\mathcal{G}, A)$. Combining

Assumption 2 with Lemma 1, we obtain $c_i > 0$. Moreover, it is easy to obtain that

$$\begin{aligned} \mathcal{L}V_i(e_i(t), t) &= 2e_i(t) \left[-e_i(t) + \sum_{j=1}^n [\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))] + u_i(t) \right] \\ &\quad + |g_i(y_i(t), t) - g_i(x_i(t), t)|^2 \\ &\leq -2e_i^2(t) + 2|e_i(t)| \sum_{j=1}^n |\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))| \\ &\quad + 2e_i(t)u_i(t) + |g_i(y_i(t), t) - g_i(x_i(t), t)|^2. \end{aligned}$$

Next, we will prove the following inequality:

$$|\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))| \leq A_{ij}\rho_j|e_j(t)|. \quad (8)$$

In fact, according to Assumption 1, we will discuss from the following three situations.

1) For $|y_j(t)| \leq T_j$, $|x_j(t)| \leq T_j$, one has

$$\begin{aligned} |\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))| &\leq a_{ij}^*|f_j(y_j(t)) - f_j(x_j(t))| \\ &\leq qA_{ij}\rho_j|y_j(t) - x_j(t)|. \end{aligned}$$

2) For $|y_j(t)| > T_j$, $|x_j(t)| > T_j$, we obtain that

$$\begin{aligned} |\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))| &\leq a_{ij}^{**}|f_j(y_j(t)) - f_j(x_j(t))| \\ &\leq A_{ij}\rho_j|y_j(t) - x_j(t)|. \end{aligned}$$

3) For $|x_j(t)| \leq T_j < |y_j(t)|$ or $|y_j(t)| \leq T_j < |x_j(t)|$, here, we only consider $|x_j(t)| \leq T_j < |y_j(t)|$, because the other case can be discussed in a similar way. Then, we have $y_j(t) < -T_j \leq x_j(t)$ or $x_j(t) \leq T_j < y_j(t)$. Here, we only consider $T_j < y_j(t)$ or $y_j(t) < -T_j$, because other cases can be similarly discussed as above.

If $T_j < y_j(t)$, we have

$$\begin{aligned} |\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))| &\leq |\hat{a}_{ij}(t)(f_j(y_j(t)) - f_j(T_j))| + |\bar{a}_{ij}(t)(f_j(x_j(t)) - f_j(T_j))| \\ &\leq A_{ij}|f_j(y_j(t)) - f_j(T_j)| + A_{ij}|f_j(x_j(t)) - f_j(T_j)| \\ &\leq A_{ij}\rho_j|y_j(t) - x_j(t)|. \end{aligned}$$

If $y_j(t) < -T_j$, we obtain that

$$\begin{aligned} |\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))| &\leq |\hat{a}_{ij}(t)(f_j(x_j(t)) - f_j(-T_j))| \\ &\quad + |\bar{a}_{ij}(t)(f_j(-T_j) - f_j(y_j(t)))| \\ &\leq A_{ij}|f_j(x_j(t)) - f_j(-T_j)| + A_{ij}|f_j(-T_j) - f_j(y_j(t))| \\ &\leq A_{ij}\rho_j|y_j(t) - x_j(t)|. \end{aligned}$$

So, combined with the above situations 1)–3), we can conclude that (8) holds. Hence, by (8), we obtain

$$\begin{aligned} 2e_i(t) \sum_{j=1}^n [\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))] &\leq 2 \sum_{j=1}^n A_{ij}\rho_j|e_i(t)||e_j(t)| \end{aligned}$$

$$\leq 2 \sum_{j=1}^n A_{ij} \rho_j e_i^2(t) + \sum_{j=1}^n A_{ij} \rho_j [e_j^2(t) - e_i^2(t)].$$

When $t \in [mT, mT + \delta]$, we have

$$\begin{aligned} 2e_i(t)u_i(t) &\leq -2d_i e_i(t)[e_i(\delta(t)) - e_i(t)] - 2d_i e_i^2(t) \\ &\leq 2\lambda_i e_i^2(t) + \frac{d_i^2}{2\lambda_i} |e_i(t) - e_i(\delta(t))|^2 - 2d_i e_i^2(t). \end{aligned}$$

Therefore, in the light of the definition of function $V(e(t), t)$, we can derive

$$\begin{aligned} \mathbb{E} \mathcal{L}V(e(t), t) &= \mathbb{E} \sum_{i=1}^n c_i \mathcal{L}V_i(e_i(t), t) \\ &\leq \sum_{i=1}^n c_i \left(-2 + 2 \sum_{j=1}^n A_{ij} \rho_j + 2\lambda_i - 2d_i + \eta_i^2 \right) \mathbb{E} e_i^2(t) \\ &\quad + \mathbb{E} \sum_{i=1}^n c_i \sum_{j=1}^n A_{ij} \rho_j [e_j^2(t) - e_i^2(t)] \\ &\quad + \sum_{i=1}^n c_i \frac{d_i^2}{2\lambda_i} \mathbb{E} |e_i(t) - e_i(\delta(t))|^2. \end{aligned}$$

Meanwhile, according to Assumption 2 and Lemma 1, we obtain

$$\sum_{i=1}^n c_i \sum_{j=1}^n A_{ij} \rho_j [e_j^2(t) - e_i^2(t)] = 0 \quad (9)$$

and then

$$\begin{aligned} \mathbb{E} \mathcal{L}V(e(t), t) &\leq \sum_{i=1}^n c_i \left(-2 + 2 \sum_{j=1}^n A_{ij} \rho_j + 2\lambda_i - 2d_i + \eta_i^2 \right) \mathbb{E} e_i^2(t) \\ &\quad + \sum_{i=1}^n c_i \frac{d_i^2}{2\lambda_i} \mathbb{E} |e_i(t) - e_i(\delta(t))|^2. \end{aligned} \quad (10)$$

On the basis of system (6), we obtain

$$\begin{aligned} &\mathbb{E} |e_i(t) - e_i(\delta(t))|^2 \\ &\leq 2 \mathbb{E} \int_{\delta(t)}^t \left[\tau \left| -e_i(s) + \sum_{j=1}^n [\hat{a}_{ij}(s)f_j(y_j(s)) - \bar{a}_{ij}(s)f_j(x_j(s))] \right. \right. \\ &\quad \left. \left. - d_i e_i(s) \right|^2 + |g_i(y_i(s), s) - g_i(x_i(s), s)|^2 \right] ds \\ &\leq 2 \mathbb{E} \int_{\delta(t)}^t \left[3\tau \left(e_i^2(s) + \left| \sum_{j=1}^n \hat{a}_{ij}(s)f_j(y_j(s)) - \bar{a}_{ij}(s)f_j(x_j(s)) \right|^2 \right. \right. \\ &\quad \left. \left. + d_i^2 e_i^2(s) \right) + \eta_i^2 e_i^2(s) \right] ds. \end{aligned} \quad (11)$$

According to (8), one has

$$\begin{aligned} &\left| \sum_{j=1}^n \hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t)) \right|^2 \\ &\leq n \sum_{j=1}^n |\hat{a}_{ij}(t)f_j(y_j(t)) - \bar{a}_{ij}(t)f_j(x_j(t))|^2 \end{aligned}$$

$$\begin{aligned} &\leq n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{j=1}^n A_{ij} \rho_j e_i^2(t) \\ &\quad + n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{j=1}^n A_{ij} \rho_j [e_j^2(t) - e_i^2(t)]. \end{aligned} \quad (12)$$

Combining (11) with (12), we can easily derive

$$\begin{aligned} &\mathbb{E} |e_i(t) - e_i(\delta(t))|^2 \\ &\leq 2 \int_{\delta(t)}^t \left[3\tau \left(1 + n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{j=1}^n A_{ij} \rho_j \right) + \eta_i^2 \right] \mathbb{E} e_i^2(s) ds \\ &\quad + \int_{\delta(t)}^t 6\tau n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{j=1}^n A_{ij} \rho_j \mathbb{E} [e_j^2(s) - e_i^2(s)] ds \\ &\quad + 6\tau^2 d_i^2 \mathbb{E} e_i^2(\delta(t)). \end{aligned}$$

Moreover, we can also obtain that

$$\begin{aligned} &\sum_{i=1}^n c_i \mathbb{E} |e_i(t) - e_i(\delta(t))|^2 \\ &\leq \int_{\delta(t)}^t \sum_{i=1}^n c_i \left[3\tau \left(1 + n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{j=1}^n A_{ij} \rho_j \right) + \eta_i^2 \right] \mathbb{E} e_i^2(s) ds \\ &\quad + \sum_{i=1}^n c_i 6\tau^2 d_i^2 \mathbb{E} e_i^2(\delta(t)) \\ &\quad + \int_{\delta(t)}^t 6\tau n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{i=1}^n \sum_{j=1}^n c_i A_{ij} \rho_j \mathbb{E} [e_j^2(s) - e_i^2(s)] ds. \end{aligned}$$

According to (9), we have

$$\begin{aligned} &\int_{\delta(t)}^t 6\tau n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{i=1}^n \sum_{j=1}^n c_i A_{ij} \rho_j \\ &\quad [e_j^2(s) - e_i^2(s)] ds = 0. \end{aligned}$$

Therefore, we obtain

$$\begin{aligned} &\sum_{i=1}^n c_i \mathbb{E} |e_i(t) - e_i(\delta(t))|^2 \\ &\leq \int_{\delta(t)}^t \sum_{i=1}^n c_i \left[3\tau \left(1 + n \max_{i,j \in \mathcal{N}} \{A_{ij} \rho_j\} \sum_{j=1}^n A_{ij} \rho_j \right) + \eta_i^2 \right] \mathbb{E} e_i^2(s) ds \\ &\quad + \sum_{i=1}^n c_i 6\tau^2 d_i^2 \mathbb{E} e_i^2(\delta(t)) \\ &\leq \int_{\delta(t)}^t \phi \sum_{i=1}^n c_i \mathbb{E} |e_i(s) - e_i(\delta(s))|^2 ds \\ &\quad + \sum_{i=1}^n c_i (\tau \phi_i + 6\tau^2 d_i^2) \mathbb{E} e_i^2(\delta(t)). \end{aligned} \quad (13)$$

According to Lemma 2, we have

$$\begin{aligned} &\sum_{i=1}^n c_i \mathbb{E} |e_i(t) - e_i(\delta(t))|^2 \\ &\leq \exp\{\phi \tau\} \sum_{i=1}^n c_i (\tau \phi_i + 6\tau^2 d_i^2) \mathbb{E} e_i^2(\delta(t)) \end{aligned}$$

$$\begin{aligned} &\leq 2 \exp\{\phi\tau\} \sum_{i=1}^n c_i \left(\tau\phi_i + 6\tau^2 d_i^2 \right) \mathbb{E}|e_i(t) - e_i(\delta(t))|^2 \\ &\quad + 2 \exp\{\phi\tau\} \sum_{i=1}^n c_i \left(\tau\phi_i + 6\tau^2 d_i^2 \right) \mathbb{E}e_i^2(t). \end{aligned} \quad (14)$$

Therefore, combining (13) with (14), we have

$$\sum_{i=1}^n c_i \mathbb{E}|e_i(t) - e_i(\delta(t))|^2 \leq \frac{K(\tau)}{1-K(\tau)} \sum_{i=1}^n c_i \mathbb{E}e_i^2(t). \quad (15)$$

Then, substituting (15) into (10), we obtain

$$\begin{aligned} \mathbb{E}\mathcal{L}V(e(t), t) &\leq \sum_{i=1}^n c_i \left(-2 + 2 \sum_{j=1}^n A_{ij}\rho_j + 2\lambda_i \right. \\ &\quad \left. - 2d_i + \eta_i^2 + \frac{dK(\tau)}{1-K(\tau)} \right) \mathbb{E}e_i^2(t) \\ &= - \sum_{i=1}^n c_i \theta_i \mathbb{E}e_i^2(t) = -\beta \mathbb{E}V(e(t), t). \end{aligned} \quad (16)$$

Meanwhile, we know $\mathbb{E}\mathcal{L}V(e(t), t) = D^+ \mathbb{E}V(e(t), t)$. Therefore, combined with (16), we can obtain

$$\mathbb{E}V(e(t), t) \leq V(e(mT), mT) \exp\{-\beta(t - mT)\}. \quad (17)$$

When $t \in (mT + \delta, (m+1)T)$, we obtain

$$\begin{aligned} \mathbb{E}\mathcal{L}V(e(t), t) &\leq \mathbb{E} \sum_{i=1}^n c_i \left(-2e_i^2(t) + 2 \sum_{j=1}^n A_{ij}\rho_j e_i^2(t) \right. \\ &\quad \left. + \eta_i^2 e_i^2(t) + \sum_{j=1}^n A_{ij}\rho_j [e_j^2(t) - e_i^2(t)] \right). \end{aligned}$$

According to (9), we obtain

$$\sum_{i=1}^n c_i \sum_{j=1}^n A_{ij}\rho_j [e_j^2(t) - e_i^2(t)] = 0$$

and then

$$\begin{aligned} \mathbb{E}\mathcal{L}V(e(t), t) &\leq \mathbb{E} \sum_{i=1}^n c_i \left(-2 + \sum_{j=1}^n A_{ij}\rho_j + \eta_i^2 \right) e_i^2(t) \\ &\leq \max_{i \in \mathcal{N}} \left\{ \eta_i^2 + \sum_{j=1}^n A_{ij}\rho_j - 2 \right\} \mathbb{E}V(e(t), t) \\ &= \alpha \mathbb{E}V(e(t), t). \end{aligned}$$

Hence, we obtain

$$\mathbb{E}V(e(t), t) \leq V(e(mT + \delta), mT + \delta) \exp\{\alpha(t - mT - \delta)\}$$

and

$$\begin{aligned} \mathbb{E}V(e(mT + \delta), mT + \delta) &\leq V(e(mT + \delta), mT + \delta) \\ &\quad \times \exp\{\alpha(t - mT - \delta)\}. \end{aligned} \quad (18)$$

According to (17) and (18), it can be observed that when $0 \leq t \leq \delta$

$$\mathbb{E}V(e(t), t) \leq V(e_0, 0) \exp\{-\beta t\}$$

and

$$\mathbb{E}V(e(\delta), \delta) \leq V(e_0, 0) \exp\{-\beta\delta\}.$$

When $\delta < t < T$, it can be easily obtained that

$$\begin{aligned} \mathbb{E}V(e(t), t) &\leq V(e(\delta), \delta) \exp\{\alpha(t - \delta)\} \\ &\leq V(e_0, 0) \exp\{-\beta\delta + \alpha(t - \delta)\} \end{aligned}$$

and

$$\begin{aligned} \mathbb{E}V(e(T), T) &\leq V(e(\delta), \delta) \exp\{\alpha(T - \delta)\} \\ &\leq V(e_0, 0) \exp\{-\beta\delta + \alpha(T - \delta)\}. \end{aligned}$$

When $T \leq t \leq T + \delta$, it follows:

$$\begin{aligned} \mathbb{E}V(e(t), t) &\leq V(e(T), T) \exp\{-\beta(t - T)\} \\ &\leq V(e_0, 0) \exp\{-\beta\delta - \beta(t - T) + \alpha(T - \delta)\} \end{aligned}$$

and

$$\mathbb{E}V(e(T + \delta), T + \delta) \leq V(e_0, 0) \exp\{-2\beta\delta + \alpha(T - \delta)\}.$$

When $T + \delta < t < 2T$, we derive

$$\begin{aligned} \mathbb{E}V(e(t), t) &\leq V(e_0, 0) \exp\{-2\beta\delta + \alpha(T - \delta) + \alpha(t - T - \delta)\} \\ &\text{and} \end{aligned}$$

$$\mathbb{E}V(e(2T), 2T) \leq V(e_0, 0) \exp\{-2\beta\delta + \alpha(T - \delta)\}.$$

According to the induction, it can be derived that when $mT \leq t \leq mT + \delta$, that is, $(t - \delta)/T \leq m \leq t/T$

$$\begin{aligned} \mathbb{E}V(e(t), t) &\leq \mathbb{E}V(e(mT), mT) \exp\{-\beta(t - mT)\} \\ &\leq V(e_0, 0) \exp\{-m\beta\delta + m\alpha(T - \delta)\} \\ &\leq V(e_0, 0) \exp\left\{-\frac{\delta\beta - (T - \delta)\alpha}{T}(t - \delta)\right\} \end{aligned}$$

and

$$\begin{aligned} \mathbb{E}V(e(t), t) &\leq \mathbb{E}V(e(mT + \delta), mT + \delta) \exp\{\alpha(t - mT - \delta)\} \\ &\leq V(e_0, 0) \exp\{(-m - 1)\beta\delta + (m + 1)\alpha(T - \delta)\} \\ &\leq V(e_0, 0) \exp\left\{-\frac{\delta\beta - (T - \delta)\alpha}{T}(t - \delta)\right\}. \end{aligned}$$

Therefore, for any $t > 0$, we can show that

$$\begin{aligned} \mathbb{E}V(e(t), t) &\leq V(e_0, 0) \exp\left\{-\frac{\delta\beta - (T - \delta)\alpha}{T}(t - \delta)\right\} \\ &\leq \max_{i \in \mathcal{N}} \{c_i\} |e_0|^2 \exp\left\{-\frac{\delta\beta - (T - \delta)\alpha}{T}(t - \delta)\right\}. \end{aligned}$$

In addition, we know that $V(e(t), t) \geq \min_{i \in \mathcal{N}} \{c_i\} |e(t)|^2$. Hence, one has

$$\mathbb{E}|e(t)|^2 \leq \frac{\max_{i \in \mathcal{N}} \{c_i\}}{\min_{i \in \mathcal{N}} \{c_i\}} |e_0|^2 \exp\{-r(t - \delta)\}$$

where $r = (\delta\beta - (T - \delta)\alpha)/T$. By means of Definition 2, we can derive that drive system (1) and response system (2) achieve exponential synchronization under controller (3). The proof is completed. ■

Remark 5: Let $\hat{\theta} = \delta/T$ be the control rate, and $T^* = (\delta/T)\beta - (1 - (\delta/T))\alpha = \hat{\theta}\beta - (1 - \hat{\theta})\alpha = \hat{\theta}(\alpha + \beta) - \alpha$ can be the convergence rate of exponential synchronization. The smaller $\hat{\theta}$ is, the smaller T^* is. So the slower stochastic neural

networks converge. Moreover, when we fix other variables and the convergence rate T^* , if the control strength d_i is large, the control rate $\hat{\theta} = \delta/T$ can be relatively small.

The above discussions are about the exponential synchronization of stochastic neural networks via controller (3). Next, we will consider a special case. When $\delta \rightarrow T$, controller (3) becomes FCDSO, that is to say

$$u_i(t) = -d_i(y_i(\delta(t)) - x_i(\delta(t))), \quad i \in \mathcal{N}. \quad (19)$$

Under controller (19), we will provide some corollaries about the synchronization of stochastic neural networks as follows.

Corollary 1: Under Assumptions 1 and 2, there exists a sufficiently small duration τ between two consecutive state observations such that $k(\tau) < 1$. Then, drive system (1) and response system (2) reach synchronization in mean square under controller (19) if the following inequalities are satisfied:

$$\alpha > 0, \quad \beta > 0.$$

Proof: Set $V(e, t) = \sum_{i=1}^n c_i V_i(e_i, t)$, where $V_i(e_i, t) = e_i^2$. From the proof of Theorem 1, we can derive that

$$\mathbb{E} \mathcal{L} V(e(t), t) \leq -\beta \mathbb{E} V(e(t), t).$$

Therefore, one has

$$\mathbb{E} V(e(t), t) \leq C - \beta \int_0^t \sum_{i=1}^n c_i \mathbb{E} e_i^2(s) ds$$

then it yields

$$\int_0^t \mathbb{E} |e(s)|^2 ds \leq \frac{C}{D}$$

where $C = \mathbb{E} V(e_0, 0)$ and $D = \beta \min_{i \in \mathcal{N}} \{c_i\}$. So when $t \rightarrow \infty$, we have

$$\int_0^\infty \mathbb{E} |e(t)|^2 dt < \infty.$$

Hence, by Definition 3, we know drive system (1) and response system (2) reach synchronization in mean square under controller (19). This completes the proof. ■

Remark 6: Generally, we cannot obtain $\lim_{t \rightarrow \infty} \mathbb{E} |e(t)|^2 = 0$ from $\int_0^\infty \mathbb{E} |e(t)|^2 dt = 0$. For example, we know that $\int_0^\infty \sin x^2 dx < \infty$, however, $\lim_{x \rightarrow \infty} \sin x^2 \neq 0$. In fact, if the condition “ $\mathbb{E} |e(t)|^2$ is uniformly continuous” is added in Definition 3, we can obtain $\lim_{t \rightarrow \infty} \mathbb{E} |e(t)|^2 = 0$, which means asymptotically synchronous in mean square is more difficult to achieve. However, in this paper, a corollary is given to show that asymptotically synchronous in mean square can be achieved. At the same time, compared with asymptotically synchronous in mean square, exponential synchronization is more difficult to achieve.

Corollary 2: Under Assumptions 1 and 2, there exists a sufficiently small duration τ between two consecutive state observations such that $k(\tau) < 1$. Then, drive system (1) and response system (2) reach asymptotical synchronization in mean square under controller (19) if the following inequalities are satisfied:

$$\alpha > 0, \quad \beta > 0.$$

Proof: We can easily obtain

$$\begin{aligned} \mathbb{E} e_i^2(t) &= e_i^2(0) \\ &+ \mathbb{E} \int_0^t \left\{ 2e_i(v) \left[-e_i(v) + \sum_{j=1}^n [\hat{a}_{ij}(v) f_j(y_j(v)) \right. \right. \\ &\quad \left. \left. - \bar{a}_{ij}(v) f_j(x_j(v)) \right] - d_i e_i(\delta(v)) \right\} \\ &\quad \left. + |g_i(y_i(v), v) - g_i(x_i(v), v)|^2 \right\} dv. \end{aligned}$$

According to Assumptions 1 and 2, for any $t \geq 0$, we have

$$\begin{aligned} \mathbb{E} e_i^2(t) &\leq e_i^2(0) + G \int_0^t \mathbb{E} e_i^2(v) dv + G \int_0^t \mathbb{E} |e(v)|^2 dv \\ &\quad + G \int_0^t \mathbb{E} |e_i(v) - e_i(\delta(v))|^2 dv \end{aligned} \quad (20)$$

in which G represents a variable positive constant. For any $v \geq 0$, there exists a unique integer $s \geq 0$, such that $v \in [s\tau, (s+1)\tau)$. So, for any $\xi \in [s\tau, v]$, we have $\delta(\xi) = s\tau$.

Moreover, according to the error system (6), we can obtain

$$\begin{aligned} \mathbb{E} |e_i(v) - e_i(\delta(v))|^2 &\leq 2 \int_{\delta(v)}^v (3\tau + \eta_i^2) \mathbb{E} e_i^2(\xi) d\xi \\ &\quad + 2n \max_{i,j \in \mathcal{N}} \{A_{ij}^2 \rho_j^2\} \int_{\delta(v)}^v 3\tau \mathbb{E} |e(\xi)|^2 d\xi \\ &\quad + 12\tau^2 d_i^2 [\mathbb{E} e_i^2(v) + \mathbb{E} |e_i(v) - e_i(\delta(v))|^2]. \end{aligned}$$

According to $K(\tau) < 1$, which yields that $12\tau^2 d_i^2 < 1$, we can obtain

$$\begin{aligned} \mathbb{E} |e_i(v) - e_i(\delta(v))|^2 &\leq \frac{2(3\tau + \eta_i^2)}{1 - 12\tau^2 d_i^2} \int_{\delta(v)}^v \mathbb{E} e_i^2(\xi) d\xi \\ &\quad + \frac{6\tau \max_{i,j \in \mathcal{N}} \{A_{ij}^2 \rho_j^2\}}{1 - 12\tau^2 d_i^2} \int_{\delta(v)}^v \mathbb{E} |e(\xi)|^2 d\xi \\ &\quad + \frac{12\tau^2 d_i^2}{1 - 12\tau^2 d_i^2} \mathbb{E} e_i^2(v). \end{aligned} \quad (21)$$

Substituting (21) into (20), we have

$$\begin{aligned} \mathbb{E} e_i^2(t) &\leq \mathbb{E} e_i^2(0) + G \int_0^t \mathbb{E} e_i^2(v) dv + G \int_0^t \mathbb{E} |e(v)|^2 dv \\ &\quad + G \int_0^t \int_{\delta(v)}^v e_i^2(\xi) d\xi dv + G \int_0^t \int_{\delta(v)}^v \mathbb{E} |e(\xi)|^2 d\xi dv. \end{aligned}$$

For $v - \tau \leq \delta(v)$, it shows

$$\begin{aligned} \int_0^t \int_{\delta(v)}^v \mathbb{E} e_i^2(\xi) d\xi dv &\leq \int_0^t \int_{v-\tau}^v \mathbb{E} e_i^2(\xi) d\xi dv \\ &\leq \tau \int_{-\tau}^t \mathbb{E} e_i^2(\xi) d\xi. \end{aligned}$$

Similarly, it yields

$$\int_0^t \int_{\delta(v)}^v \mathbb{E} |e(\xi)|^2 d\xi dv \leq \tau \int_{-\tau}^t \mathbb{E} |e(\xi)|^2 d\xi.$$

Therefore, we can easily obtain

$$\mathbb{E}|e(t)|^2 \leq \mathbb{E}|e_0|^2 + G \int_0^t \mathbb{E}e_i^2(v)dv + G \int_0^t \mathbb{E}|e(v)|^2 dv.$$

According to Corollary 1, we know that $G \int_0^t \mathbb{E}e_i^2(v)dv$ and $G \int_0^t \mathbb{E}|e(v)|^2 dv$ are both bounded. Therefore, we obtain

$$\mathbb{E}e_i^2(t) \leq G.$$

And since

$$\begin{aligned} & \sum_{i=1}^n \left(\mathbb{E}e_i^2(t_2) - \mathbb{E}e_i^2(t_1) \right) \\ &= \sum_{i=1}^n \mathbb{E} \int_{t_1}^{t_2} \left\{ 2e_i(t) \left[-e_i(t) + \sum_{j=1}^n [\hat{a}_{ij}(t)f_j(y_j(t)) \right. \right. \\ & \quad \left. \left. - \bar{a}_{ij}(t)f_j(x_j(t))] - d_i e_i(\delta(t)) \right] \right. \\ & \quad \left. + |g_i(y_i(t), t) - g_i(x_i(t), t)|^2 \right\} dt \end{aligned}$$

combining it with Assumption 1, we obtain

$$\mathbb{E}|e(t_2)|^2 - \mathbb{E}|e(t_1)|^2 \leq G(t_2 - t_1), \text{ for } 0 \leq t_1 < t_2 < \infty.$$

Hence, $\mathbb{E}|e(t)|^2$ is uniformly continuous. Then, combined with $\int_0^\infty \mathbb{E}|e(t)|^2 dt < \infty$, we obtain $\lim_{t \rightarrow \infty} \mathbb{E}|e(t)|^2 = 0$. According to Definition 4, we know that drive system (1) and response system (2) are asymptotically synchronous in mean square under controller (19). Then, the proof is completed. ■

In the following text, we will show an exponential synchronization criterion about stochastic neural networks under controller (19).

Theorem 2: Under Assumptions 1 and 2, suppose there exists a sufficiently small τ , such that $K(\tau) < 1$, then drive system (1) and response system (2) reach exponential synchronization under controller (19) if the following inequalities are satisfied:

$$\alpha > 0, \beta > 0.$$

Proof: Let Lyapunov function $V(e(t), t) = \sum_{i=1}^n c_i V_i(e_i(t), t)$, where $V_i(e_i(t), t) = e_i^2(t)$. Since $V(e(t), t) \geq c^*|e(t)|^2$, where $c^* = \min_{i \in \mathcal{N}}\{c_i\}$, by using the Itô's formula and (16), we have

$$\begin{aligned} & \exp\{\gamma t\} c^* \mathbb{E}|e(t)|^2 \\ & \leq V(e_0, 0) + \int_0^t \exp\{\gamma \xi\} [\gamma \mathbb{E}V(e(\xi), \xi) + \mathbb{E}\mathcal{L}V(e(\xi), \xi)] d\xi \\ & \leq V(e_0, 0) \\ & \quad + \int_0^t \exp\{\gamma \xi\} \left[\gamma \mathbb{E}V(e(\xi), \xi) - \sum_{i=1}^n c_i \theta_i \mathbb{E}e_i^2(\xi) \right] d\xi \\ & \leq V(e_0, 0) \end{aligned}$$

where γ is a positive constant and we let $\gamma < \min_{i \in \mathcal{N}}\{\theta_i\}$. Hence, we obtain $\mathbb{E}|e(t)|^2 \leq [V(e_0, 0)/c^*] \exp\{-\gamma t\}$. From Definition 2, we know drive system (1) and response

system (2) reaches exponential synchronization under controller (19). Then, the proof is completed. ■

Remark 7: In a few decades, many researchers have investigated the dynamical properties of stochastic neural networks. For example, in [26], the globally exponential stability problem for a class of discrete-time stochastic memristive neural networks was studied. Different from the previous work, this paper uses FCDSO which is more appropriate in practice instead of feedback control based on the continuous-time state observations.

Remark 8: In recent years, many researchers have used all sorts of controllers to investigate the dynamical properties of neural networks. For example, Wu *et al.* [48] used the sampled-data feedback controller to study the global exponential synchronization problem for chaotic neural networks. And by employing periodically alternate output feedback control, the global exponential synchronization problem of two memristive chaotic recurrent neural networks with time-varying delays was discussed in [49]. When PIDOC is reduced to FCDSO, we found that few researchers utilize FCDSO to investigate the exponential synchronization of stochastic neural networks. FDCSO and PIDOC are used to study the synchronization of stochastic neural networks in this paper.

IV. NUMERICAL EXAMPLE

In this section, a numerical example is given to show the effectiveness of our theoretical results.

Consider the following stochastic neural networks:

$$\begin{aligned} dx_i(t) = & \left(-x_i(t) + \sum_{j=1}^3 a_{ij}(x_j(t))f_j(x_j(t)) + I_i \right) dt \\ & + g_i(x_i(t), t)dB(t), \quad t \geq 0, i = 1, 2, 3 \end{aligned} \quad (22)$$

and

$$\begin{aligned} dy_i(t) = & \left(-y_i(t) + \sum_{j=1}^3 a_{ij}(y_j(t))f_j(y_j(t)) + I_i + u_i(t) \right) dt \\ & + g_i(y_i(t), t)dB(t), \quad t \geq 0, i = 1, 2, 3 \end{aligned} \quad (23)$$

where $I_1 = I_2 = I_3 = 0.0002$. And $a_{11}(x_1(t)) = a_{22}(x_2(t)) = a_{33}(x_3(t)) = 0$, $g_i(x_i, t) = 0.1x_i$

$$\begin{aligned} a_{21}(x_1(t)) &= \begin{cases} 1, & |x_2(t)| \leq 1 \\ 1.6, & |x_2(t)| > 1 \end{cases} \\ a_{12}(x_2(t)) &= \begin{cases} 0, & |x_2(t)| \leq 1 \\ 6.6, & |x_2(t)| > 1 \end{cases} \\ a_{31}(x_1(t)) &= \begin{cases} 6.5, & |x_1(t)| \leq 1 \\ 1.3, & |x_1(t)| > 1 \end{cases} \\ a_{13}(x_3(t)) &= \begin{cases} 1.25, & |x_3(t)| \leq 1 \\ 2, & |x_3(t)| > 1 \end{cases} \\ a_{23}(x_3(t)) &= \begin{cases} 2.2, & |x_3(t)| \leq 1 \\ 1.5, & |x_3(t)| > 1 \end{cases} \\ a_{32}(x_2(t)) &= \begin{cases} 0.1, & |x_2(t)| \leq 1 \\ -5, & |x_2(t)| > 1. \end{cases} \end{aligned}$$

And take the activation functions as

$$f_1(x_1(t)) = \sin(|x_1(t)| - 1), \quad f_2(x_2(t)) = \tanh(|x_2(t)| - 1)$$

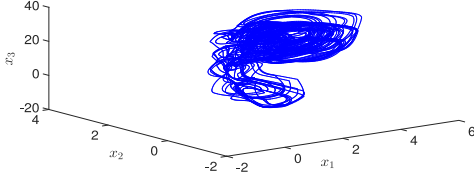
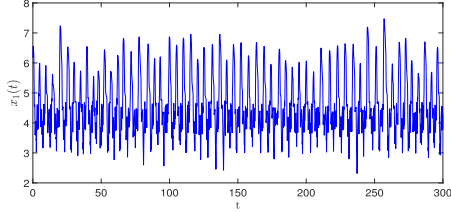
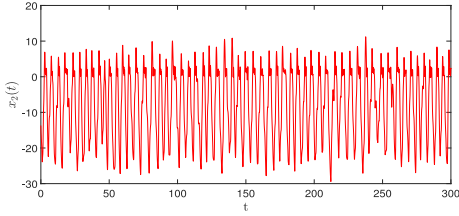


Fig. 2. Chaotic attractor of system (22).

Fig. 3. State $x_1(t)$ of system (22).Fig. 4. State $x_2(t)$ of system (22).

$$\begin{aligned} f_3(x_3(t)) &= \sin(|x_3(t)| - 1), \quad f_1(y_1(t)) = \sin(|y_1(t)| - 1) \\ f_2(y_2(t)) &= \tanh(|y_2(t)| - 1), \quad f_3(y_3(t)) = \sin(|y_3(t)| - 1). \end{aligned}$$

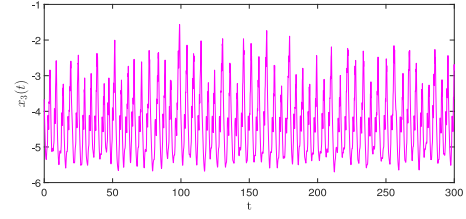
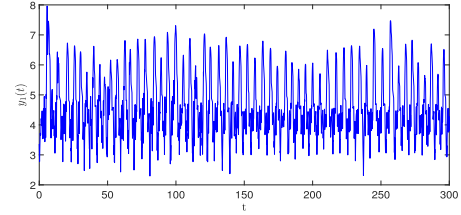
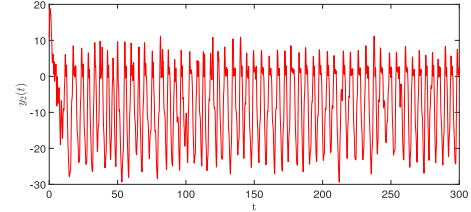
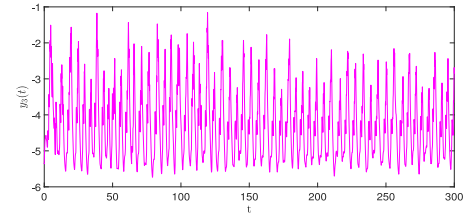
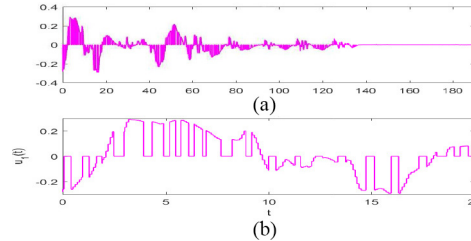
Moreover, system (22) can be considered as a drive system, and system (23) is the corresponding response system. Now, we define the synchronization error vector as follows: $e_i(t) = y_i(t) - x_i(t)$ for $i = 1, 2, 3$. Combining (22) with (23), we obtain

$$\begin{aligned} de_i(t) &= \left(-e_i(t) + \sum_{j=1}^3 \hat{a}_{ij}(t)f_j(y_j(t)) - \sum_{j=1}^3 \bar{a}_{ij}(t)f_j(x_j(t)) + u_i(t) \right) dt \\ &\quad + (g_i(y_i(t), t) - g_i(x_i(t), t))dB(t) \\ i &= 1, 2, 3. \end{aligned} \quad (24)$$

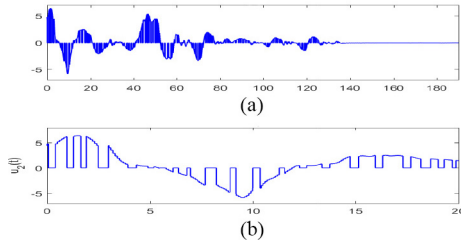
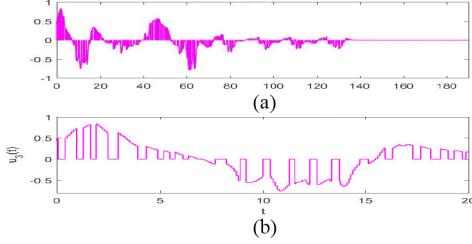
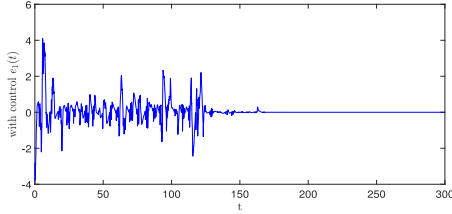
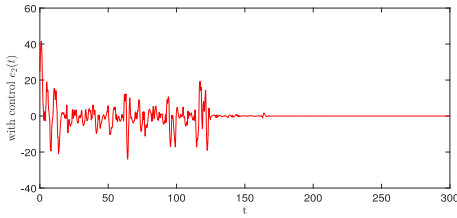
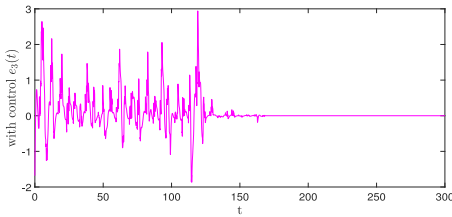
Meanwhile, system (22) has a chaotic attractor with the initial value that $x_1(0) = 1.5$, $x_2(0) = 0.6$, and $x_3(0) = 1.4$ (see Fig. 2). We can see that Figs. 3–5 depict the state variables $x_1(t)$, $x_2(t)$, and $x_3(t)$ of system (22), respectively.

By a simple computation, we obtain $A_{11} = 0$, $A_{12} = 6.6$, $A_{13} = 2$, $A_{21} = 35$, $A_{22} = 0$, $A_{23} = 2.2$, $A_{31} = 6.5$, $A_{32} = 0.1$, and $A_{33} = 0$, where $i, j = 1, 2, 3$. Then, we set $T = 1$, $\delta = 0.5$, $\tau = 0.001$, $\eta_1 = \eta_2 = \eta_3 = 0.1$, $\lambda_1 = \lambda_2 = \lambda_3 = 0.5$, $d_1 = d_2 = d_3 = 30$, $\rho_j^- = -0.2$, and $\rho_j^+ = 0.2$ for $j = 1, 2, 3$ and we obtain $\phi = 0.9458$, $d = 900$, $K(\tau) = 0.0254$, $\alpha = 5.3100$, and $\beta = 21.9274$.

It is easy to check that all conditions of Theorem 1 are satisfied. Figs. 6–8 show the state variables $y_1(t)$, $y_2(t)$, and $y_3(t)$ of system (23), respectively. Figs. 9–11 show

Fig. 5. State $x_3(t)$ of system (22).Fig. 6. State $y_1(t)$ of system (23) under controller $u_1(t)$.Fig. 7. State $y_2(t)$ of system (23) under controller $u_2(t)$.Fig. 8. State $y_3(t)$ of system (23) under controller $u_3(t)$.Fig. 9. State of controller $u_1(t)$.

the controllers $u_1(t)$, $u_2(t)$, and $u_3(t)$, respectively. Finally, Figs. 12–14 show the synchronization error vector of the state variables $e_1(t)$, $e_2(t)$, and $e_3(t)$ between drive system (22) and response system (23) under the controllers $u_1(t)$, $u_2(t)$, and $u_3(t)$, respectively. Therefore, drive system (22) and response system (23) realize exponential synchronization.

Fig. 10. State of controller $u_2(t)$.Fig. 11. State of controller $u_3(t)$.Fig. 12. Synchronization error vector $e_1(t)$.Fig. 13. Synchronization error vector $e_2(t)$.Fig. 14. Synchronization error vector $e_3(t)$.

V. CONCLUSION

In this paper, the exponential synchronization of stochastic neural networks was investigated. A new control strategy was proposed to analyze the exponential synchronization of stochastic neural networks. In addition, our control degenerates to a common FCDSO when the control width tends to the control period, which was also discussed in this paper.

Furthermore, some corollaries were obtained about synchronization in mean square, asymptotical synchronization in mean square, and exponential synchronization of stochastic neural networks under FCDSO. In our future work, we will consider the following topics.

- 1) We will use aperiodically intermittent pinning discrete observation control to study the exponential synchronization of stochastic neural networks.
- 2) We will consider the case that the duration τ between two consecutive observations becomes time-varying $\tau(t)$.
- 3) We will continue to extend the control considered in this paper to the nonperiodic case.

ACKNOWLEDGMENT

The authors would like to thank the editors and reviewers for their valuable comments.

REFERENCES

- [1] K. Mathiyalagan, R. Anbuviya, R. Sakthivel, J. H. Park, and P. Prakash, "Non-fragile H_∞ synchronization of memristor-based neural networks using passivity theory," *Neural Netw.*, vol. 74, pp. 85–100, Feb. 2016.
- [2] K. Shi, J. Wang, Y. Tang, and S. Zhong, "Reliable asynchronous sampled-data filtering of T-S fuzzy uncertain delayed neural networks with stochastic switched topologies," *Fuzzy Sets Syst.*, to be published. doi: [10.1016/j.fss.2018.11.017](https://doi.org/10.1016/j.fss.2018.11.017).
- [3] X. Yang and D. W. C. Ho, "Synchronization of delayed memristive neural networks: Robust analysis approach," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3377–3387, Dec. 2016.
- [4] Q. Xiao and Z. Zeng, "Scale-limited Lagrange stability and finite-time synchronization for memristive recurrent neural networks on time scales," *IEEE Trans. Cybern.*, vol. 47, no. 10, pp. 2984–2994, Oct. 2017.
- [5] R. Zhang, D. Zeng, J. H. Park, S. Zhong, and Y. Yu, "Novel discontinuous control for exponential synchronization of memristive recurrent neural networks with heterogeneous time-varying delays," *J. Frankl. Inst. Eng. Appl. Math.*, vol. 355, no. 5, pp. 2826–2848, 2018.
- [6] H. Bao, J. Cao, J. Kurths, A. Alsaedi, and B. Ahmad, " H_∞ state estimation of stochastic memristor-based neural networks with time-varying delays," *Neural Netw.*, vol. 99, pp. 79–91, Mar. 2018.
- [7] H. Wang, S. Duan, T. Huang, L. Wang, and C. Li, "Exponential stability of complex-valued memristive recurrent neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 28, no. 3, pp. 766–771, Mar. 2017.
- [8] W. Zhang, X. Yang, C. Xu, J. Feng, and C. Li, "Finite-time synchronization of discontinuous neural networks with delays and mismatched parameters," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3761–3771, Aug. 2018.
- [9] W. Zhang, S. Yang, C. Li, W. Zhang, and X. Yang, "Stochastic exponential synchronization of memristive neural networks with time-varying delays via quantized control," *Neural Netw.*, vol. 104, pp. 93–103, Aug. 2018.
- [10] J. Cheng, J. Park, J. Cao, and W. Qi, "Hidden Markov model-based nonfragile state estimation of switched neural network with probabilistic quantized outputs," *IEEE Trans. Cybern.*, to be published. doi: [10.1109/TCYB.2019.2909748](https://doi.org/10.1109/TCYB.2019.2909748).
- [11] S. Li, H. Su, and X. Ding, "Synchronized stationary distribution of hybrid stochastic coupled systems with applications to coupled oscillators and a Chua's circuits network," *J. Frankl. Inst. Eng. Appl. Math.*, vol. 355, no. 17, pp. 8743–8765, 2018.
- [12] A. Abdurahman and H. Jiang, "New results on exponential synchronization of memristor-based neural networks with discontinuous neuron activations," *Neural Netw.*, vol. 84, pp. 161–171, Dec. 2016.
- [13] H. Zhang, F. Yang, X. Liu, and Q. Zhang, "Stability analysis for neural networks with time-varying delay based on quadratic convex combination," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 24, no. 4, pp. 513–521, Apr. 2013.
- [14] J.-L. Wang, H.-N. Wu, T. Huang, S.-Y. Ren, J. Wu, and X.-X. Zhang, "Analysis and control of output synchronization in directed and undirected complex dynamical networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 8, pp. 3326–3338, Aug. 2018.

- [15] J.-L. Wang, Z. Qin, H.-N. Wu, T. Huang, and P.-C. Wei, "Analysis and pinning control for output synchronization and $\mathcal{H}_\infty$ output synchronization of multiweighted complex networks," *IEEE Trans. Cybern.*, vol. 49, no. 4, pp. 1314–1326, Apr. 2019.
- [16] Y. Liu, D. Zhang, J. Lou, J. Lu, and J. Cao, "Stability analysis of quaternion-valued neural networks: Decomposition and direct approaches," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4201–4211, Sep. 2018.
- [17] X. Lv, X. Li, J. Cao, and M. Perc, "Dynamical and static multisynchronization of coupled multistable neural networks via impulsive control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 12, pp. 6062–6072, Dec. 2018.
- [18] Y. Liu, W. Li, and J. Feng, "The stability of stochastic coupled systems with time-varying coupling and general topology structure," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4189–4200, Sep. 2018.
- [19] Z. Guo, J. Wang, and Z. Yan, "Global exponential synchronization of two memristor-based recurrent neural networks with time delays via static or dynamic coupling," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 45, no. 2, pp. 235–249, Feb. 2015.
- [20] S. Yang, C. Li, and T. Huang, "Impulsive synchronization for TS fuzzy model of memristor-based chaotic systems with parameter mismatches," *Int. J. Control Autom. Syst.*, vol. 14, no. 3, pp. 854–864, 2016.
- [21] J. Cheng, C. Ahn, H. Karimi, J. Cao, and W. Qi, "An event-based asynchronous approach to Markov jump systems with hidden mode detections and missing measurements," *IEEE Trans. Syst., Man, Cybern., Syst.*, to be published. doi: [10.1109/TSMC.2018.2866906](https://doi.org/10.1109/TSMC.2018.2866906).
- [22] K. Shi, Y. Tang, S. Zhong, C. Yin, X. Huang, and W. Wang, "Nonfragile asynchronous control for uncertain chaotic Lurie network systems with Bernoulli stochastic process," *Int. J. Robust Nonlinear Control*, vol. 28, no. 5, pp. 1693–1714, 2018.
- [23] G. Zhang and Y. Shen, "Exponential synchronization of delayed memristor-based chaotic neural networks via periodically intermittent control," *Neural Netw.*, vol. 55, pp. 1–10, Jul. 2014.
- [24] Z. Tang, J. Park, Y. Wang, and J. Feng, "Distributed impulsive quasi-synchronization of Lur'e networks with proportional delay," *IEEE Trans. Cybern.*, vol. 49, no. 8, pp. 3105–3115, Aug. 2019.
- [25] G. Zhang, Y. Shen, and L. Wang, "Global anti-synchronization of a class of chaotic memristive neural networks with time-varying delays," *Neural Netw.*, vol. 46, pp. 1–8, Oct. 2013.
- [26] H. Liu, Z. Wang, B. Shen, T. Huang, and F. E. Alsaadi, "Stability analysis for discrete-time stochastic memristive neural networks with both leakage and probabilistic delays," *Neural Netw.*, vol. 102, pp. 1–9, Jun. 2018.
- [27] J. Wang, K. Shi, Q. Huang, S. Zhong, and D. Zhang, "Stochastic switched sampled-data control for synchronization of delayed chaotic neural networks with packet dropout," *Appl. Math. Comput.*, vol. 335, pp. 211–230, Oct. 2018.
- [28] K. Shi, J. Wang, S. Zhong, X. Zhang, Y. Liu, and J. Cheng, "New reliable nonuniform sampling control for uncertain chaotic neural networks under Markov switching topologies," *Appl. Math. Comput.*, vol. 347, pp. 169–193, Apr. 2019.
- [29] S. Ding, Z. Wang, and H. Zhang, "Quasi-synchronization of delayed memristive neural networks via region-partitioning-dependent intermittent control," *IEEE Trans. Cybern.*, to be published.
- [30] Y. Xu, H. Zhou, and W. Li, "Stabilisation of stochastic delayed systems with Levy noise on networks via periodically intermittent control," *Int. J. Control*, to be published. doi: [10.1080/00207179.2018.1479538](https://doi.org/10.1080/00207179.2018.1479538).
- [31] P. Wang, W. Jin, and H. Su, "Synchronization of coupled stochastic complex-valued dynamical networks with time-varying delays via aperiodically intermittent adaptive control," *Chaos*, vol. 28, no. 4, 2018, Art. no. 043114.
- [32] P. Wang, B. Zhang, and H. Su, "Stabilization of stochastic uncertain complex-valued delayed networks via aperiodically intermittent nonlinear control," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 49, no. 3, pp. 649–662, Mar. 2019.
- [33] H. Zhou and W. Li, "Synchronisation of stochastic-coupled intermittent control systems with delays and Levy noise on networks without strong connectedness," *IET Control Theory Appl.*, vol. 13, no. 1, pp. 36–49, 2019.
- [34] C. Xu, X. Yang, J. Lu, J. Feng, F. Alsaadi, and T. Hayat, "Finite-time synchronization of networks via quantized intermittent pinning control," *IEEE Trans. Cybern.*, vol. 48, no. 10, pp. 3021–3027, Oct. 2018.
- [35] Y. Wu, H. Li, and W. Li, "Intermittent control strategy for synchronization analysis of time-varying complex dynamical networks," *IEEE Trans. Syst., Man, Cybern., Syst.*, to be published. doi: [10.1109/TSMC.2019.2920451](https://doi.org/10.1109/TSMC.2019.2920451).
- [36] S. Yang, C. Li, and T. Huang, "Exponential stabilization and synchronization for fuzzy model of memristive neural networks by periodically intermittent control," *Neural Netw.*, vol. 75, pp. 162–172, Mar. 2016.
- [37] X. Mao, "Stabilization of continuous-time hybrid stochastic differential equations by discrete-time feedback control," *Automatica*, vol. 49, no. 12, pp. 3677–3681, 2013.
- [38] S. You, W. Liu, J. Lu, X. Mao, and Q. Qiu, "Stabilization of hybrid systems by feedback control based on discrete-time state observations," *SIAM J. Control Optim.*, vol. 53, no. 2, pp. 905–925, 2015.
- [39] S. You, L. Hu, W. Mao, and X. Mao, "Robustly exponential stabilization of hybrid uncertain systems by feedback controls based on discrete-time observations," *Stat. Probab. Lett.*, vol. 102, pp. 8–16, Jul. 2015.
- [40] Y. Wu, B. Chen, and W. Li, "Synchronization of stochastic coupled systems via feedback control based on discrete-time state observations," *Nonlinear Anal. Hybrid Syst.*, vol. 26, pp. 68–85, Nov. 2017.
- [41] A. F. Filippov, *Differential Equations With Discontinuous Right-Hand Side*. Dordrecht, The Netherlands: Kluwer, 1988.
- [42] L. O. Chua and L. Yang, "Cellular neural networks: Applications," *IEEE Trans. Circuits Syst.*, vol. 35, no. 10, pp. 1273–1290, Oct. 1988.
- [43] J. J. Hopfield, "Neurons with graded response have collective computational properties like those of two-state neurons," *Proc. Nat. Acad. Sci. USA*, vol. 81, no. 10, pp. 3088–3092, 1984.
- [44] M. Y. Li and Z. Shuai, "Global-stability problem for coupled systems of differential equations on networks," *J. Differ. Equ.*, vol. 248, no. 1, pp. 1–20, 2010.
- [45] X. Mao, *Stochastic Differential Equations and Applications*. Chichester, U.K.: Horwood, 1997.
- [46] Y. Yang, W. Zhou, P. Shi, X. Yang, X. Zhou, and H. Su, "Synchronization of delayed neural networks with Levy noise and Markovian switching via sampled data," *Nonlinear Dyn.*, vol. 81, no. 3, pp. 1179–1189, 2015.
- [47] Y. Xu, Z. He, and P. Wang, "pth moment asymptotic stability for neutral stochastic functional differential equations with Levy processes," *Appl. Math. Comput.*, vol. 269, pp. 594–605, Oct. 2015.
- [48] H. Wu, R. Li, H. Wei, X. Zhang, and R. Yao, "Synchronization of a class of memristive neural networks with time delays via sampled-data control," *Int. J. Mach. Learn. Cybern.*, vol. 6, no. 3, pp. 365–373, 2015.
- [49] X. Li, J.-A. Fang, and H. Lin, "Exponential synchronization of memristive chaotic recurrent neural networks via alternate output feedback control," *Asian J. Control*, vol. 20, no. 1, pp. 469–482, 2018.



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